Library Management System with Book Recommendations

An intelligent platform for library operations through personalised discovery

MSC COMPUTER SCIENCE

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PROJECT OVERVIEW

Transforming Library Experience

Purpose

Digitise library operations whilst enhancing user discovery through intelligent book recommendations aligned with interests and reading history

Key Objectives

- User-friendly book browsing interface
- Intelligent recommendation engine
- Streamlined digital workflows
- Data-driven insights dashboard
- Secure, scalable architecture

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USER ROLES

Three-Tier Access System

Student

- Browse & search catalogue
- Receive personalised recommendations
- Request books digitally
- Track borrowing history
- Manage profile preferences

Librarian

- Department-specific dashboard
- Approve/reject borrow requests
- Access student profiles
- View analytics for their department
- Monitor popular books

Administrator

- Full system access across departments
- Complete book management (CRUD)
- User account management
- System-wide analytics
- Global oversight

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ARCHITECTURE

Modern Full-Stack Technology

Backend

Django 6.0.1 with REST Framework

JWT authentication, SQLite/PostgreSQL, NumPy & Pandas for recommendations

Frontend

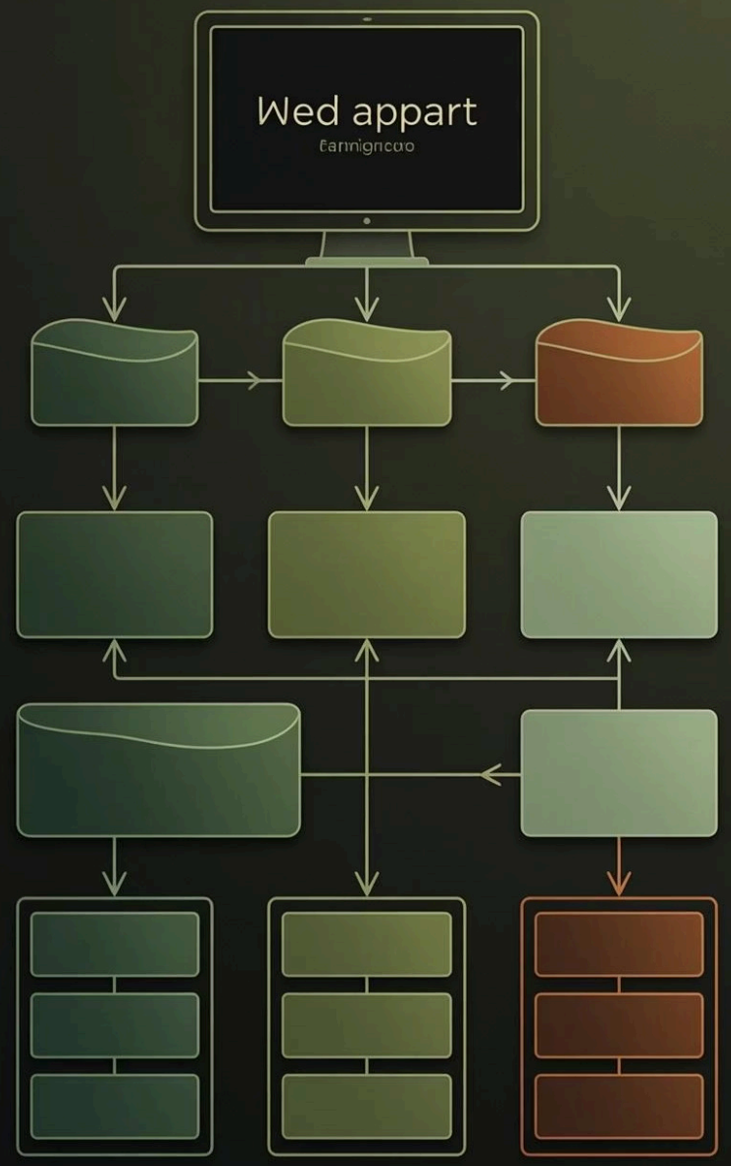
React 19.2.0 with Vite

Tailwind CSS, Framer Motion, Recharts visualisations, component-based design

Security

JWT tokens & RBAC

CORS protection, input validation, secure password hashing, XSS prevention



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INTELLIGENT FEATURES

Triple-Strategy Recommendation Engine

01

Content-Based Filtering

Analyses preferred categories from user profile, ideal for new users with limited history

02

Interaction-Based Filtering

Leverages browsing and borrowing patterns, excludes previously interacted books

03

Hybrid Approach

Combines both methods for balanced, personalised recommendations (default strategy)

Powered by NumPy & Pandas: Advanced data processing delivers relevant suggestions based on user preferences, reading history, and similar reader behaviour

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BORROWING SYSTEM

Streamlined Digital Workflow

1 Student Requests

Browse catalogue, select book, submit digital request

2 Request Queued

Status: "requested", routed to appropriate librarian

3 Librarian Reviews

Department-based access, approve or reject with stock checking

4 Approved & Tracked

Quantity decrements, student tracks in "My Borrows"

Safety Features

- Prevents duplicate active requests
- Real-time stock availability checking
- Transaction-safe quantity updates
- Department-based access control
- Complete borrowing history tracking
- Status notifications throughout

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DATABASE DESIGN

Entity Relationship Architecture

Core Entities

- **User & UserProfile**

  Custom authentication with student profiles, department links, preferences

- **Book Catalogue**

  Title, authors, categories, ratings, quantity, thumbnails, metadata

- **BookInteraction**

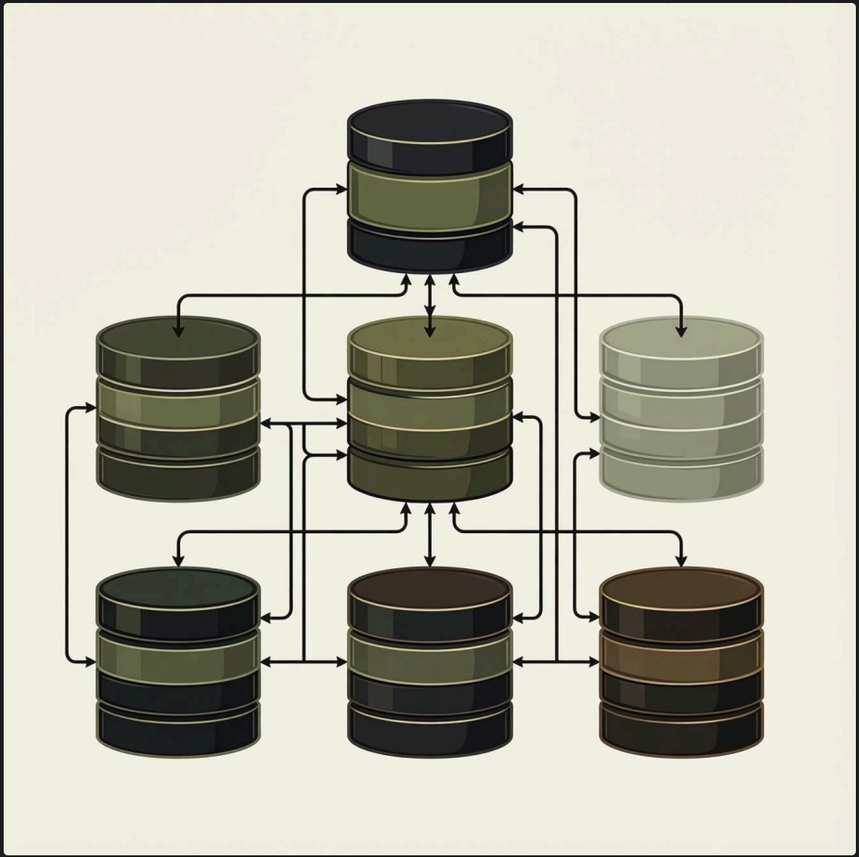
  Tracks views, likes, borrows for recommendation engine

- **Borrow Records**

  Complete lifecycle: requested, approved, returned with timestamps

- **Department**

  Organises students and librarians by academic department



Many-to-many relationships managed via BookInteraction and Borrow tables, enabling rich analytics

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ANALYTICS DASHBOARD

Data-Driven Insights

3

Statistic Categories

Books, students, borrows

30

Day Activity Window

Recent active students tracking

100%

Department Coverage

Librarians & admins

Book Statistics

- Total books in catalogue
- In-stock vs out-of-stock
- Most viewed titles
- Most borrowed books
- Top categories

Student Metrics

- Total registered students
- Active students (overall)
- Active last 30 days
- Department distribution
- Reading patterns

Borrow Analysis

- Requested count
- Approved transactions
- Returned books
- Popular borrowing times
- Department trends

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IMPLEMENTATION HIGHLIGHTS

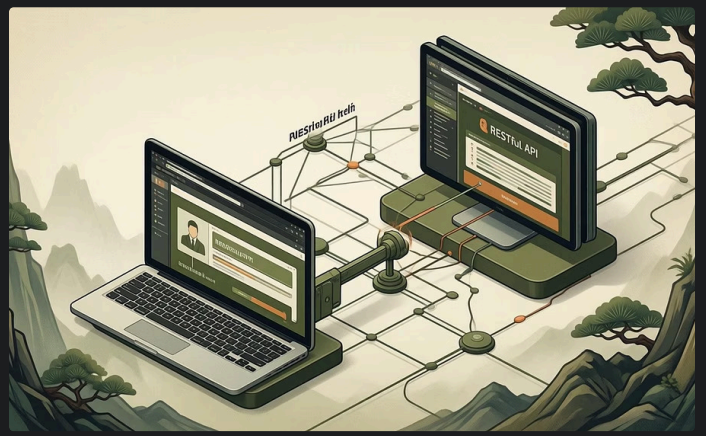
Technical Excellence

Responsive Design

Mobile-friendly with Tailwind CSS, smooth Framer Motion animations, optimised layouts

JWT Authentication

60-minute access tokens, 24-hour refresh tokens, role-based access control



RESTful API

Clean endpoint structure, comprehensive serializers, automatic token refresh

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> **Student Onboarding:** New students encounter an interest selector modal to choose preferred categories, immediately improving recommendation accuracy

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✓ CONCLUSION

Successful Modern Library Platform

Project Achievements

- Full-stack application with intelligent features
- Triple-strategy recommendation engine
- Role-based access across three user types
- Comprehensive analytics dashboard
- Secure, scalable architecture

Future Enhancements

Machine learning models, collaborative filtering, book reviews & ratings, email notifications, predictive analytics

Production-ready foundation for real-world library management with appropriate hardening and feature expansion

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